STPM Physics · Semester 3 · Chapter 21

# WAVE MOTION

## 21.1 Progressive Waves

### DEFINITIONS & LAWS

- A **progressive (travelling) wave** is a wave that transfers energy from one point to another through a medium (or vacuum) by oscillations of the particles (or fields), without transferring matter.
- **Wavelength** is the distance between two successive points that are in phase on the wave (the distance travelled by the wave in one period).
- **Amplitude** is the maximum displacement of a particle from its equilibrium position; **period** is the time for one complete oscillation of a particle.
- **Wave speed** is the distance travelled by the wavefront per unit time.
- **Phase difference** between two points on a progressive wave depends on their separation along the direction of travel.
- In a transverse wave the oscillations are perpendicular to the direction of propagation; in a longitudinal wave they are parallel to it.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave speed** | $v = f\lambda$ | $v$ = wave speed (m $s^{-1}$); $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Wave travelling in +x direction** | $y = A\sin(\omega t - kx)$ | $y$ = displacement (m); $A$ = amplitude (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad $m^{-1}$); $\omega = 2\pi f$ (rad $s^{-1}$) |
| **Wave speed in terms of ω and k** | $v = \frac{\omega}{k} = \frac{\lambda}{T}; f = \frac{\omega}{2\pi}$ | $v$ = wave speed (m $s^{-1}$); $\omega$ = angular frequency (rad $s^{-1}$); $k$ = wave number (rad $m^{-1}$); $T$ = period (s); $f$ = frequency (Hz) |
| **Phase difference** | $\varphi = \frac{2\pi}{\lambda}\Delta x; \varphi = \frac{2\pi\Delta t}{T}$ | $\varphi$ = phase difference (rad); $\Delta x$ = path separation (m); $\Delta t$ = time difference (s); $T$ = period (s) |
| **Wave travelling in −x direction** | $y = A\sin(\omega t + kx)$ | Travels in the −x direction; symbols as above |

### TIPS TO ANSWER

- Read ω and k from the equation by comparison with $y = A\sin(\omega t - kx)$; then $v = \frac{\omega}{k}$ and $f = \frac{\omega}{2\pi}$.

- Phase difference: write $\varphi = \frac{2\pi\Delta x}{\lambda}$, substitute, and give the answer in radian (and degree if asked) with 3 s.f.
- For graphs: y against x gives λ and A; y against t gives T and A. Label axes with units.
- Say that energy is transferred in the direction of wave propagation but the particles only oscillate about their equilibrium positions.

### SECTION A TRAPS

**Trap:** Using $v = \omega k$ instead of $\frac{\omega}{k}$. Check units: $\frac{\omega}{k}$ gives m $s^{-1}$.

**Trap:** Taking the number multiplying t as f. In $y = A\sin(50t - 4.0x)$, 50 is ω (rad $s^{-1}$), so $f = \frac{50}{2\pi} \approx 8.0$ Hz.

**Trap:** Phase difference without 2π: $\varphi = \frac{\Delta x}{\lambda}$ is the fraction of a cycle, not the phase angle in radian.

**Trap:** Reading $y = A\sin[2\pi(\frac{t}{T} - \frac{x}{\lambda})]$ with the denominators swapped; $\frac{t}{T}$ gives period and $\frac{x}{\lambda}$ gives wavelength.

### TRICKS

**Trick:** Two points separated by one wavelength are in phase (2π); separated by $\frac{\lambda}{2}$ they are in antiphase (π); by $\frac{\lambda}{4}$ the phase difference is $\frac{\pi}{2}$.

**Trick:** In the form $\sin[2\pi(\frac{t}{T} - \frac{x}{\lambda})]$, read T and λ directly and take $v = \frac{\lambda}{T}$.

## 21.2 Wave Intensity

### DEFINITIONS & LAWS

- **Intensity** of a wave is the rate of energy transferred per unit area perpendicular to the direction of propagation (power per unit area), unit W $m^{-2}$.
- Intensity is directly proportional to the square of the amplitude (for the same frequency and medium).
- For a point source radiating uniformly in all directions with no absorption, intensity is inversely proportional to the square of the distance from the source (inverse square law).
- Explain the inverse square law: the same power spreads over the surface of a sphere of area $4\pi r^2$, which increases as $r^2$.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Intensity** | $I = \frac{P}{A}$ | I = intensity (W $m^{-2}$); P = power (W); A = area normal to the wave ($m^2$) |
| **Intensity of a point source** | $I = \frac{P}{4\pi r^2}$ | Point source; r = distance from the source (m); P = power (W) |
| **Intensity and amplitude** | $I \propto A^2$; $\frac{I_2}{I_1} = (\frac{A_2}{A_1})^2$ | A = amplitude; I = intensity (W $m^{-2}$) |
| **Inverse square law** | $I_1r_1^2 = I_2r_2^2$; $I_2 = I_1(\frac{r_1}{r_2})^2$ | $I_1$, $I_2$ = intensities at distances $r_1$, $r_2$ from the source (W $m^{-2}$, m) |
| **Amplitude of a spherical wave** | $A \propto \frac{1}{r}$ | Spherical wave; A = amplitude; r = distance from the source (m) |

## TIPS TO ANSWER

- Choose between $I \propto \frac{1}{r^2}$ (distance change) and $I \propto A^2$ (amplitude change), and write the relation before substituting.
- Show the ratio: $I_2 = I_1(\frac{r_1}{r_2})^2$ with values and unit, then the answer to 3 s.f. in W $m^{-2}$.
- State the assumptions: point source, spreads uniformly, no absorption by the medium.
- For power calculations, use area $4\pi r^2$ and give the answer with unit W $m^{-2}$.

## SECTION A TRAPS

**Trap:** Using $I \propto \frac{1}{r}$ instead of $\frac{1}{r^2}$. Doubling the distance gives one quarter, not one half.

**Trap:** Inverting the ratio: moving farther must decrease the intensity, so the factor $(\frac{r_1}{r_2})^2$ is less than 1.

**Trap:** Tripling the amplitude triples the intensity. Wrong: $I \propto A^2$, so the intensity becomes 9 times.

**Trap:** Using the diameter as r in $I = \frac{P}{4\pi r^2}$.

## TRICKS

**Trick:** Distance × 3 → intensity ÷ 9; amplitude × 3 → intensity × 9.

**Trick:** Intensity down by a factor of 4 means distance up by a factor of 2: $r_2 = r_1\sqrt{\frac{I_1}{I_2}}$.

# 21.3 Principle of Superposition

## DEFINITIONS & LAWS

- **Principle of superposition**: when two or more waves meet at a point, the resultant displacement is the vector sum of the displacements of the individual waves at that point.
- **Coherent sources** are sources that emit waves of the same frequency with a constant phase difference.
- **Constructive interference** occurs when the waves meet in phase: the path difference is $n\lambda$ ($n = 0, 1, 2, \ldots$), giving the maximum resultant amplitude.
- **Destructive interference** occurs when the waves meet in antiphase: the path difference is $(n + ½)\lambda$, giving the minimum resultant amplitude.
- A stable, observable interference pattern requires coherent waves of similar amplitude and the same frequency.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Constructive interference** | $\Delta x = n\lambda$ | $\Delta x$ = path difference (m); $\lambda$ = wavelength (m); $n = 0, 1, 2, \ldots$; coherent in-phase sources |
| **Destructive interference** | $\Delta x = (n + ½)\lambda$ | $\Delta x$ = path difference (m); $\lambda$ = wavelength (m); $n = 0, 1, 2, \ldots$; coherent in-phase sources |
| **Phase difference** | $\varphi = \frac{2\pi}{\lambda}\Delta x$ | $\varphi$ = phase difference (rad); $\Delta x$ = path difference (m); $\lambda$ = wavelength (m) |
| **Resultant amplitude** | $A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\varphi$ | Two waves of amplitudes $A_1$ and $A_2$ (m); $\varphi$ = phase difference (rad) |
| **Intensity of equal waves** | $I = 4I_0\cos^2\frac{\varphi}{2}$ | Equal waves; $I_0$ = intensity of each wave (W m$^{-2}$); $\varphi$ = phase difference (rad) |
| **Special cases of phase difference** | $\varphi = \frac{\pi}{2}$: $A = \sqrt{A_1^2 + A_2^2}$; $\varphi = 0$: $A = A_1 + A_2$; $\varphi = \pi$: $A = \lvert A_1 - A_2\rvert$ | A = resultant amplitude (m); $A_1$, $A_2$ = amplitudes of the two waves (m); $\varphi$ in rad |

## TIPS TO ANSWER

- Compute $\frac{\Delta x}{\lambda}$ and state the result: a whole number gives constructive; an odd number of half-wavelengths gives destructive.
- Define coherence completely: same frequency and constant phase difference (not just 'same amplitude').
- For the amplitude of the resultant wave, state φ, then substitute into the amplitude relation with unit.
- Compare the intensity with $I_0$ using $I \propto A^2$, and state the answer in terms of $I_0$.

### SECTION A TRAPS

**Trap:** Concluding destructive interference just because the path difference is 'not an integer'. Check whether $\frac{\Delta x}{\lambda}$ is a half-integer, e.g. 1.5.

**Trap:** Simply adding amplitudes (3 + 4 = 7) for a phase difference of $\frac{\pi}{2}$; the correct result is $\sqrt{3^2 + 4^2}$ = 5.

**Trap:** Converting the path difference to phase without 2π: $\varphi = \frac{2\pi\Delta x}{\lambda}$, so $\Delta x = \frac{\lambda}{3}$ gives $\frac{2\pi}{3}$ rad.

**Trap:** Saying that waves of different frequencies give a stable pattern; a constant phase difference requires the same frequency.

### TRICKS

**Trick:** $\frac{\Delta x}{\lambda}$ = 3.0, 1.5, 0.75: integer → bright (constructive); half-integer → dark (destructive); anything else is in between.

**Trick:** $I = 4I_0\cos^2\frac{\varphi}{2}$: $\varphi = \frac{2\pi}{3}$ gives $I_0$; φ = π gives 0; φ = 0 gives $4I_0$.

## 21.4 Standing Waves

### DEFINITIONS & LAWS

- A **standing (stationary) wave** is formed by the superposition of two progressive waves of the same frequency and amplitude travelling in opposite directions.
- A **node** is a point of zero amplitude (permanent destructive interference); an **antinode** is a point of maximum amplitude.

- Adjacent nodes (or adjacent antinodes) are separated by $\frac{\lambda}{2}$; a node and the adjacent antinode are separated by $\frac{\lambda}{4}$.
- Differences from a progressive wave: a standing wave does not transfer energy along the medium; its amplitude varies with position (zero at nodes); particles between two adjacent nodes oscillate in phase, and particles in adjacent segments oscillate in antiphase.
- For a string fixed at both ends there is a node at each end; only wavelengths with an integral number of half-wavelengths fit on the string.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Standing wave on string fixed at both ends** | $L = \frac{n\lambda}{2}$; $f_n = \frac{nv}{2L}$ | n = 1, 2, 3, ...; v = wave speed on the string (m $s^{-1}$); L = length (m); λ = wavelength (m); $f_n$ = frequency of the nth mode (Hz) |
| **Fundamental and harmonics** | $f_1 = \frac{v}{2L}$; $f_n = nf_1$ | $f_1$ = fundamental frequency (Hz); $f_n$ = frequency of the nth harmonic (Hz); v = wave speed (m $s^{-1}$); L = length (m) |
| **Standing wave equation** | $y = 2A \sin(kx) \cos(\omega t)$ | $k = \frac{2\pi}{\lambda}$ (rad $m^{-1}$); A = amplitude of each travelling wave (m); ω = angular frequency (rad $s^{-1}$); nodes at $kx = n\pi$ |
| **Node spacing** | Distance between adjacent nodes = $\frac{\lambda}{2} = \frac{\pi}{k}$ | λ = wavelength (m); $k = \frac{2\pi}{\lambda}$ = wave number (rad $m^{-1}$) |
| **Wave speed on a string** | $v = \sqrt{\frac{T}{\mu}}$ | Stretched string; T = tension (N); μ = mass per unit length (kg $m^{-1}$) |

## TIPS TO ANSWER

- Sketch the pattern with N at both ends for a string and label the nodes, antinodes and the wavelength on the diagram.
- State the condition $L = \frac{n\lambda}{2}$, then $f = \frac{v}{\lambda} = \frac{nv}{2L}$; give the answer to 3 s.f. in Hz.
- In an 'explain' question state: formed by superposition of the incident and reflected waves of the same frequency and amplitude travelling in opposite directions.
- From $y = 2A \sin(kx) \cos(\omega t)$ identify k, then $\lambda = \frac{2\pi}{k}$ and the node separation $\frac{\lambda}{2}$.

## SECTION A TRAPS

**Trap:** Taking the node separation to be λ rather than $\frac{\lambda}{2}$.

**Trap:** Using $f_1 = \frac{v}{L}$ for a string fixed at both ends; the fundamental has $L = \frac{\lambda}{2}$, so $f_1 = \frac{v}{2L}$.

**Trap:** Counting the gaps between nodes wrongly: from the 2nd node to the 5th node there are three gaps, i.e. $3 \times \frac{\lambda}{2}$.

**Trap:** Saying that a standing wave transports energy along the string, or that all points have the same amplitude.

### TRICKS

**Trick:** The nth harmonic is n times the fundamental; the third harmonic of a 200 Hz fundamental is 600 Hz.

**Trick:** From the number of gaps g between two nodes, distance = $g\frac{\lambda}{2}$; from a node at 0 and k, the spacing is $\frac{\pi}{k}$.

## 21.5 Electromagnetic Waves

### DEFINITIONS & LAWS

- **Electromagnetic waves** are transverse waves consisting of oscillating electric and magnetic fields that are perpendicular to each other and to the direction of propagation; they do not need a material medium.
- All electromagnetic waves travel at the same speed, $c = 3.00\times10^{8}$ m $s^{-1}$, in vacuum; they differ in frequency and wavelength.
- In an electromagnetic wave, the electric and magnetic fields oscillate in phase.
- Order of the spectrum, from longest wavelength: radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, gamma rays.
- Electromagnetic waves can be polarised, reflected, refracted, diffracted and show interference (properties of transverse waves).

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Wave equation for EM waves** | $c = f\lambda$ | $c$ = speed of light in vacuum ($3.00\times10^{8}$ m s$^{-1}$); $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Field amplitudes** | $E_0 = cB_0$ | $E_0$ = electric field amplitude (V m$^{-1}$); $B_0$ = magnetic field amplitude (T); $c$ = speed of light (m s$^{-1}$) |
| **Speed of light** | $c = \frac{1}{\sqrt{\mu_0\varepsilon_0}}$ | $\mu_0 = 4\pi\times10^{-7}$ H m$^{-1}$; $\varepsilon_0 = 8.85\times10^{-12}$ F m$^{-1}$ |
| **Photon energy** | $E = hf = \frac{hc}{\lambda}$ | $E$ = photon energy (J); $h = 6.63\times10^{-34}$ J s; $f$ = frequency (Hz); $\lambda$ = wavelength (m) |
| **Time of travel** | $t = \frac{d}{c}$ | $t$ = time (s); $d$ = distance (m); $c$ = speed of light (m s$^{-1}$) |

## TIPS TO ANSWER

- State the nature: transverse; E and B perpendicular to each other and to the direction of propagation; in phase.
- For calculations: formula → substitution with $c = 3.00\times10^{8}$ m s$^{-1}$ → answer to 3 s.f. with unit and in standard form.
- For the spectrum, order by wavelength or frequency and state the direction, e.g. 'increasing frequency'.
- State that no medium is required, so EM waves travel through vacuum.

## SECTION A TRAPS

**Trap:** Reversing the order of the spectrum: radio waves have the longest wavelength and lowest frequency; X-rays are shorter than ultraviolet.

**Trap:** Using $B = cE$ instead of $B = \frac{E}{c}$, which gives an absurd $10^{11}$ T; B is of the order of $10^{-6}$ T for E of a few hundred V m$^{-1}$.

**Trap:** Saying that E and B are 90° out of phase. They are in phase (both reach maxima together).

**Trap:** Saying that EM waves are longitudinal or that they need a medium.

## TRICKS

**Trick:** Quick check: $\lambda = \frac{c}{f} = 3\times\frac{10^{8}}{1.5}\times10^{14} = 2\times10^{-6}$ m; always examine the power of ten against the spectrum.

> **Trick:** $\frac{E}{B}$ must equal $3\times10^{8}$ m $s^{-1}$ in vacuum; use it to check magnitudes in options.